# Answers: Introduction to radians

Ifan Howells-Baines, Mark Toner

**Summary**

Answers to the questions relating to the guide on radians.

*These are the answers to Questions: Introduction to radians.*

**Please attempt the questions before reading these answers!**

## Q1

1.1. Multiplying $30^\circ$ by $\pi$ and dividing by $180$ gives $\frac{30\pi}{180}$ rad $= \frac{\pi}{6}$ rad $= 0.524$ rad to three decimal places.

1.2. Multiplying $105^\circ$ by $\pi$ and dividing by $180$ gives $\frac{105\pi}{180}$ rad $= \frac{7\pi}{12}$ rad $= 1.833$ rad to three decimal places.

1.3. Multiplying $298^\circ$ by $\pi$ and dividing by $180$ gives $\frac{298\pi}{180}$ rad $= \frac{149\pi}{90}$ rad $= 5.201$ rad to three decimal places.

1.4. Multiplying $61^\circ$ by $\pi$ and dividing by $180$ gives $\frac{61\pi}{180}$ rad $= 1.064$ rad to three decimal places.

1.5. Multiplying $353^\circ$ by $\pi$ and dividing by $180$ gives $\frac{353\pi}{180}$ rad $= 6.161$ rad to three decimal places.

1.6. Multiplying $197^\circ$ by $\pi$ and dividing by $180$ gives $\frac{197\pi}{180}$ rad $= 3.438$ rad to three decimal places.

## Q2

2.1. Multiplying $\frac{\pi}{3}$ rad by $180$ and dividing by $\pi$ gives $\frac{180\pi^\circ}{3\pi} = 60^\circ$.

2.2. Multiplying $\frac{2\pi}{3}$ rad by $180$ and dividing by $\pi$ gives $\frac{360\pi^\circ}{3\pi} = 120^\circ$.

2.3. Multiplying $\frac{\pi}{7}$ rad by $180$ and dividing by $\pi$ gives $\frac{180\pi^\circ}{7\pi} = 25.714^\circ$ to three decimal places.

2.4. Multiplying $\frac{5\pi}{7}$ rad by $180$ and dividing by $\pi$ gives $\frac{900\pi^\circ}{7\pi} = 128.571^\circ$ to three decimal places.

2.5. Multiplying $5$ rad by $180$ and dividing by $\pi$ gives $\frac{900^\circ}{\pi} = 286.479^\circ$ to three decimal places.

2.6. Multiplying $\frac{3}{4}$ rad by $180$ and dividing by $\pi$ gives $\frac{540^\circ}{4\pi} = \frac{135^\circ}{\pi} = 42.972^\circ$ to three decimal places.

## Q3

3.1. In this case, the length of the arc is $\frac{7\pi}{8} = 2.749$ (to 3dp) and the area of the sector is $\frac{49\pi}{16} = 9.621$ (to 3dp).

3.2. In this case, the length of the arc is $\frac{\pi}{2} = 1.571$ (to 3dp) and the area of the sector is $\frac{\pi}{12} = 0.262$ (to 3dp).

3.3. In this case, the length of the arc is $14\pi = 43.982$ (to 3dp) and the area of the sector is $\frac{525\pi}{2} = 824.668$ (to 3dp).

---

---

## Version history and licensing

v1.0: initial version created 08/23 by Ifan Howells-Baines, Mark Toner as part of a University of St Andrews STEP project.

- v1.1: edited 05/24 by tdhc.

This work is licensed under CC BY-NC-SA 4.0.